Uniqueness of SU(3) under Operational Distinguishability Constraints

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Abstract

We prove a classification-level result for compact connected Lie groups acting on $\mathbb{C}^3$. Under three independently motivated assumptions — a determinant constraint, the presence of a cyclic 3-symmetry, and operational distinguishability of chirally conjugate states — the only compact connected Lie group admitting a faithful irreducible unitary representation on $\mathbb{C}^3$ is SU(3). The operational distinguishability assumption is formulated via an *Information Preservation Principle* (IPP): states related by an antiunitary intertwiner must remain distinguishable within the physically admissible observable algebra, as measured by von Neumann relative entropy. We show that the IPP forces the representation to be complex ($V \not\cong \bar{V}$), eliminating SU(2) (whose 3-dimensional adjoint is real) and leaving SU(3) as the unique solution. The proof proceeds via the Peter-Weyl structure theorem, the Weyl dimension formula, and a center-quotient argument. A minimality corollary characterizes SU(3) as the minimal compact Lie group supporting a faithful irreducible unitary action on $\mathbb{C}^3$ together with a cyclic permutation symmetry and a genuinely complex representation.

Keywords: SU(3); representation theory; Peter-Weyl theorem; Weyl dimension formula; information preservation; compact Lie groups.

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1 Introduction

The classification of compact connected Lie groups and their irreducible representations is a cornerstone of modern mathematical physics. A natural question is whether specific compact Lie groups can be characterized intrinsically — that is, whether a short list of structural conditions uniquely identifies a particular group.

We address this question for SU(3). The group SU(3) plays a central role in both pure representation theory (as the simplest compact simple Lie group with a 3-dimensional complex fundamental representation) and in physics (as the gauge group of quantum chromodynamics). It is therefore of interest to identify the minimal structural conditions that force a representation to realize SU(3) rather than any other compact group.

Our main result (Theorem 1) gives three such conditions:

(1) a determinant constraint ($\rho(G) \subseteq \mathrm{SU}(3)$),

(2) the presence of a cyclic 3-symmetry (the representation contains an element conjugate to the 3-cycle permutation matrix M_3), and

(3) complex irreducibility ($V \not\cong \bar{V}$).

Under these assumptions, any compact connected Lie group G with a continuous, unitary, faithful, and irreducible representation on $\mathbb{C}^3$ is isomorphic to SU(3).

Condition (3) is the most substantive. It excludes SU(2), whose only 3-dimensional irreducible representation (the adjoint) is real. Section 3 derives condition (3) from an *Information Preservation Principle* (IPP): states related by an antiunitary intertwiner must remain operationally distinguishable. This reframes the assumption from a free postulate into a consequence of faithfulness at the level of physically admissible observables, rather than group elements alone.

The proof (Section 4) uses three standard tools: the Peter-Weyl structure theorem (to reduce to a simple factor), the tensor product decomposition of product group representations (to exclude multiple simple factors via the primeness of 3), and the Weyl dimension formula (to classify compact simple Lie groups admitting a 3-dimensional irreducible representation).

A minimality corollary (Corollary 1) characterizes SU(3) as the unique minimal compact Lie group supporting the three structural properties of Theorem 1 simultaneously.

2 Main Theorem

Theorem 1 (Uniqueness of SU(3))**.** *Let $V = \mathbb{C}^3$ with its standard Hermitian inner product. Let G be a compact, connected Lie group acting on V via a continuous, unitary, faithful, and irreducible representation $\rho : G \to \mathrm{U}(3)$. Assume:*

(1) ***(Determinant constraint)*** $\rho(G) \subseteq \mathrm{SU}(3)$.

(2) ***(3-cycle symmetry)*** *There exists $g_0 \in G$ such that $\rho(g_0)$ is unitarily equivalent to the cyclic permutation matrix*

$$M_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \tag{1}$$

(3) ***(Complex irreducibility)*** *The representation V is not isomorphic to its complex conjugate: $V \not\cong \bar{V}$.*

Then, under assumptions (1)–(3)*, $G \cong \mathrm{SU}(3)$.*

Remarks on the hypotheses

Remark 1 (Equivalent formulations)*.* Assumption (1) can be replaced by the requirement that no global U(1) scalar acts nontrivially on V, since the global U(1) factor of U(3) is exactly U(3)/SU(3). Assumption (2) is equivalent to the existence of an element $g_0 \in G$ of order 3 acting as a cyclic permutation of a basis of V. Assumption (3) is discussed in detail in Section 3.

Remark 2 (Independence of assumptions)*.* Each of (1)–(3) eliminates a distinct class of groups: SU(2) (via the adjoint) satisfies (1)–(2) but not (3); $\mathrm{U}(1)^3$ satisfies (1) and (3) but not (2) (since M_3 is not diagonal); $\mathbb{Z}_3$ satisfies (2)–(3) but not connectedness; SU(N) for $N \geq 4$ fails faithful irreducibility on $\mathbb{C}^3$. No condition is redundant.

3 The Information Preservation Principle and Complex Irreducibility

Assumption (3) of Theorem 1 appears, at first reading, as a free constraint on the representation. In this section we show that it follows from an information-theoretic faithfulness requirement on the *observable algebra* associated with V.

Definition 1 (Physically admissible observable algebra)**.** *Let $\rho : G \to \mathrm{U}(V)$ be a unitary representation. The* physically admissible observable algebra $\mathcal{A}_{\text{phys}} \subseteq \mathrm{End}(V)$ *is the set of operators corresponding to measurable quantities in the theory. It is:*

(i) a real vector space of self-adjoint operators, closed under the adjoint;

(ii) closed under conjugation by $\rho(G)$: $\mathcal{O} \in \mathcal{A}_{\text{phys}} \Rightarrow \rho(g)\,\mathcal{O}\,\rho(g)^{-1} \in \mathcal{A}_{\text{phys}}$ for all $g \in G$;

(iii) when an antiunitary intertwiner J exists (i.e. when $V \cong \bar{V}$), closed under conjugation by J: $\mathcal{O} \in \mathcal{A}_{\text{phys}} \Rightarrow J\,\mathcal{O}\,J^{-1} \in \mathcal{A}_{\text{phys}}$.

Remark 3 (Justification of closure condition (iii))*.* If J is a symmetry of the theory, it acts on states by $\psi \mapsto J\psi$ and preserves transition probabilities. Consistency of the Born rule then requires that expectation values transform equivariantly:

$$\langle\psi|\mathcal{O}|\psi\rangle = \overline{\langle J\psi|(J\mathcal{O}J^{-1})|J\psi\rangle}. \tag{2}$$

If $\mathcal{O}$ is an admissible observable, the measurement procedure for $J\psi$ must also be admissible, with corresponding operator $J\mathcal{O}J^{-1}$. Hence $\mathcal{A}_{\text{phys}}$ is closed under conjugation by J. This structure is consistent with the standard algebraic formulation of quantum theory, in which observables form a $*$-algebra acted on by symmetry automorphisms. We assume throughout that symmetries act as automorphisms of the observable algebra, as is standard in quantum theory; relaxing this would correspond to explicitly breaking the symmetry at the level of measurement, which is excluded by the hypothesis that J is a symmetry of the theory.

Proposition 1 (Information Preservation Principle)**.** *Let $\mathcal{A}_{\text{phys}}$ be the physically admissible observable algebra (Definition 1). Let $\psi_\pm = e^{\pm i\varphi}v$ for some nonzero $v \in V$ and $\varphi \neq 0$, with density matrices $\varrho_\pm = |\psi_\pm\rangle\langle\psi_\pm|$.*

Operational distinguishability: *There exists $\mathcal{O} \in \mathcal{A}_{\text{phys}}$ such that*

$$\langle\psi_+|\mathcal{O}|\psi_+\rangle \neq \langle\psi_-|\mathcal{O}|\psi_-\rangle. \tag{3}$$

The Information Preservation Principle *(IPP) requires that distinguishability persists within $\mathcal{A}_{\text{phys}}$; equivalently,*

$$D(\varrho_+\|\varrho_-) = \mathrm{Tr}(\varrho_+ \log \varrho_+ - \varrho_+ \log \varrho_-) > 0. \tag{4}$$

If the IPP holds, then $V \ncong \bar{V}$.

Proof. Suppose $V \cong \bar{V}$. By the definition of a real structure (Bröcker–tom Dieck [1], §II.6, Definition 6.1), there exists an antiunitary intertwiner $J : V \to V$ with $J\rho(g)J^{-1} = \rho(g)$ for all $g \in G$ and $\psi_- = J\psi_+$.

By hypothesis $\mathcal{A}_{\text{phys}}$ is closed under conjugation by J: for every $\mathcal{O} \in \mathcal{A}_{\text{phys}}$, $J\mathcal{O}J^{-1} \in \mathcal{A}_{\text{phys}}$. For any such $\mathcal{O}$, using antilinearity of J:

$$\langle\psi_+|\mathcal{O}|\psi_+\rangle = \overline{\langle J\psi_+|(J\mathcal{O}J^{-1})|J\psi_+\rangle} = \overline{\langle\psi_-|(J\mathcal{O}J^{-1})|\psi_-\rangle}. \tag{5}$$

Since $J\mathcal{O}J^{-1} \in \mathcal{A}_{\text{phys}}$, every observable in $\mathcal{A}_{\text{phys}}$ has a J-partner also in $\mathcal{A}_{\text{phys}}$, and Eq. (5) states that the full set of measurement outcomes on ψ_+ is determined by the corresponding J-partner outcomes on ψ_- (and conversely). The two states therefore have identical measurement statistics with respect to $\mathcal{A}_{\text{phys}}$ as a whole, contradicting the operational distinguishability (3).

Therefore $V \ncong \bar{V}$. □

Lemma 1 (Assumption (3) from IPP)**.** *Under the IPP (Proposition 1), assumption (3) of Theorem 1 is not an independent postulate: $V \ncong \bar{V}$ is a necessary consequence of requiring the physically admissible observable algebra to preserve operational distinguishability between conjugate states.*

Remark 4 (On Schur's lemma). On an irreducible representation, the G-invariant subalgebra of $\mathrm{End}(V)$ is $\mathbb{C} \cdot \mathrm{Id}$ (Schur). Distinguishability therefore cannot be witnessed by G-invariant observables alone — as is familiar in gauge theory, where physical state labels (currents, charges, helicity) are typically covariant rather than invariant. The argument above uses the full physically admissible algebra $\mathcal{A}_{\mathrm{phys}}$, not the commutant, and is therefore compatible with Schur's lemma. An explicit witness to (3) in contexts where V carries a natural Hermitian operator H is the antisymmetric part $\hat{h} = \frac{1}{2}(H - H^\top)$, which is not G-invariant but satisfies $J\hat{h}J^{-1} = -\hat{h}$ under any antiunitary intertwiner, giving $\langle\psi_+|\hat{h}|\psi_+\rangle = -\langle\psi_-|\hat{h}|\psi_-\rangle$ whenever $\langle v|\hat{h}|v\rangle \neq 0$.

4 Proof of Theorem 1

Proof of Theorem 1. Step 1 (Reduction to simple groups). By the Peter-Weyl structure theorem, G is isomorphic (up to finite cover) to

$$G \cong (G_1 \times \cdots \times G_k \times T)/\Gamma, \tag{6}$$

where each G_i is compact, connected, and simple, T is a torus, and Γ is finite. Since ρ is unitary, any torus factor acts by scalar multiplication. The determinant constraint $\rho(G) \subseteq \mathrm{SU}(3)$ (assumption (1)) eliminates nontrivial scalar actions, so the torus factor is trivial. Thus G is, up to finite cover, a product of compact simple groups.

Step 2 (Single simple factor). Suppose $G \cong G_1 \times G_2$. Every irreducible unitary representation of $G_1 \times G_2$ decomposes as $V \cong V_1 \otimes V_2$, where V_i is an irreducible unitary representation of G_i (Bröcker–tom Dieck [1], §II.4). Therefore $\dim V = (\dim V_1)(\dim V_2)$. Since $\dim V = 3$ is prime, the only factorizations over positive integers are $3 = 3 \times 1$ or $3 = 1 \times 3$, so one factor acts trivially. This contradicts faithfulness of ρ unless that factor is absent. By induction on k, the same argument applies to any product $G_1 \times \cdots \times G_k$: at most one factor acts non-trivially. Hence G is simple (up to finite cover).

Step 3 (Classification via Weyl dimension formula). Let G be compact, connected, and simple, admitting a faithful irreducible unitary representation of dimension 3. We classify such G using the Weyl dimension formula

$$\dim V_\lambda = \prod_{\alpha>0} \frac{\langle \lambda + \rho, \alpha\rangle}{\langle \rho, \alpha\rangle}, \tag{7}$$

applied to each compact simple Lie algebra. The results are summarized in Table 1.

Table 1: Minimum irreducible representation dimensions for compact simple Lie algebras.

Type	Group	Min dim	3D irrep $V \not\cong \bar{V}$?
A_1	SU(2)	2	No (real)
A_2	SU(3)	3	**Yes**
A_n, $n \geq 3$	$\mathrm{SU}(n+1)$	$n+1 \geq 4$	—
B_n	$\mathrm{SO}(2n+1)$	$2n+1 \geq 5$	—
C_n	$\mathrm{Sp}(n)$	$2n \geq 4$	—
D_n	$\mathrm{SO}(2n)$	$2n \geq 6$	—
Exceptional	$G_2, F_4, E_{6,7,8}$	≥ 7	—

Only $A_1 = \mathrm{SU}(2)$ and $A_2 = \mathrm{SU}(3)$ admit a 3-dimensional irreducible representation. SU(2)'s 3-dimensional representation (adjoint, highest weight $2\omega_1$) is *real*: $V \cong \bar{V}$, violating assumption (3). The only remaining candidate is $A_2 = \mathrm{SU}(3)$, whose fundamental representation V_{ω_1} has $\dim V_{\omega_1} = 3$ and satisfies $\mathbf{3} \not\cong \bar{\mathbf{3}}$ (the fundamental is genuinely complex).

Step 4 (Center quotients). We have $G = \mathrm{SU}(3)/Z$ for some subgroup $Z \subset Z(\mathrm{SU}(3)) \cong \mathbb{Z}_3$. The fundamental representation of SU(3) does not descend to $\mathrm{PSU}(3) = \mathrm{SU}(3)/\mathbb{Z}_3$: the

generator $\zeta = e^{2\pi i/3} \cdot I$ of $\mathbb{Z}_3$ acts as $\rho(\zeta) = \zeta \cdot I \neq I$, so $\ker \rho$ contains $\mathbb{Z}_3$ in PSU(3), making ρ non-faithful on PSU(3). Faithfulness of ρ (part of our hypotheses) therefore requires $Z = \{e\}$, giving $G = \mathrm{SU}(3)$.

Step 5 (Compatibility with M_3). M_3 is unitary and $\det(M_3) = +1$, hence $M_3 \in \mathrm{SU}(3)$. Assumption (2) is consistent with the identified group. □

Corollary 1 (Minimality of SU(3))**.** SU(3) *is the minimal compact Lie group admitting:*

(i) a faithful irreducible unitary action on $\mathbb{C}^3$,

(ii) a nontrivial cyclic permutation symmetry (an element unitarily equivalent to M_3), and

(iii) a genuinely complex representation ($V \not\cong \bar{V}$).

No proper compact subgroup of SU(3) *satisfies all three conditions simultaneously.*

Proof. By Theorem 1, SU(3) is the unique compact connected Lie group satisfying (i)–(iii). Any proper compact subgroup $H \subsetneq \mathrm{SU}(3)$ fails at least one condition: either H is finite (violating continuity), or H has no faithful irreducible 3-dimensional complex representation, since the maximal closed connected subgroups of SU(3) are U(2) and SO(3), with all other proper closed subgroups contained within these or being finite (by the classification of maximal subgroups of SU(3), see Dynkin [2]); none admit a faithful irreducible complex 3-dimensional representation. □

5 Discussion

Theorem 1 gives a minimal structural characterization of SU(3): three independently motivated conditions on a compact connected Lie group suffice to force $G \cong \mathrm{SU}(3)$. The novelty of the present formulation is Section 3, where the complex irreducibility assumption (3) is derived from an information-theoretic principle (the IPP) rather than postulated outright.

The IPP formulation clarifies what is actually required: it is not the group G itself that must distinguish chirally conjugate states, but the physically admissible observable algebra $\mathcal{A}_{\mathrm{phys}}$ (Definition 1). This sidesteps Schur's lemma and aligns with standard practice in algebraic quantum theory, where physical observables are covariant rather than G-invariant.

The minimality corollary (Corollary 1) shows that SU(3) is not merely one solution among several but the minimal compact Lie group supporting all three structural properties simultaneously. This characterization may find application in contexts where SU(3) gauge symmetry emerges from geometric or combinatorial data — for example, in lattice field theories on cubic lattices, where the 3-cycle symmetry of condition (2) is naturally present as the 120° rotational symmetry about a body diagonal. Such applications are developed in a companion paper [3].

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